SOFTWARE REQUIREMENTS SPECIFICATION

HireSphere

An AI-Powered Career Development and Recruitment Management System

Document	Software Requirements Specification (structure based on IEEE 830 / ISO 29148)
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Revision History

Version	Date	Description	Author
1.0	October 2026	Initial SRS for HireSphere (MVP and later releases).	Project team
1.1	31 October 2026	Consistency fixes (broken IDs, withdrawn-application rule, match formula, status values, data model, PDF-only CV, SSE/async AI); company hiring history and career paths; student activity streaks and points; company follow priority notifications; fake company and fake job protection; new NFR-PERF-01 and NFR-PERF-04 set; redrawn diagrams. Full list in Appendix E.	Project team

How to read changes: *table rows shaded light yellow were added or changed in version 1.1. Diagram items tagged v1.1 are new in this version.*

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Reading guide: requirement IDs use the pattern FR-<AREA>-<NN> (functional), NFR-<AREA>-<NN> (non-functional) and BR-<NN> (business rules). Priority MVP = first release; Later = deferred.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of HireSphere — An AI-Powered Career Development and Recruitment Management System. It is the reference agreement between the project stakeholders and the development team on what the system shall do, and it is the basis for design, implementation, testing and acceptance of the Minimum Viable Product (MVP) and later releases.

The intended readers are the project team and supervisors (to understand and review the requirements), developers (to design and build the system), testers (to derive test cases) and the future maintainers of the platform.

1.2 Scope

HireSphere is a web application that connects students with companies through one platform. It combines career preparation (CV analysis, skill-gap reports, practice tests, mock interviews, an AI career chatbot) with recruitment management (job posting, applications, online assessments, shortlisting, analytics). It also builds trust between the two sides through company verification, fake-job protection and transparent company hiring history. It has three portals: Student, Company and Admin.

The system consists of a React single-page frontend, a Java Spring Boot REST backend and a MySQL database, with AI features provided by an external AI service. Requirements are marked MVP (to be delivered in the first release) or Later (deferred to a subsequent release).

HireSphere does not guarantee ATS results, job eligibility, selection or placement. AI outputs are decision-support recommendations, and final hiring decisions remain with human recruiters. No automated check can prove that a company is genuine; HireSphere reduces the risk through layered checks and human review but does not guarantee that every company or job is genuine.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
ATS	Applicant Tracking System — software employers use to scan and rank CVs
API / REST	Application Programming Interface / Representational State Transfer
CV	Curriculum Vitae (resume) uploaded by a student
FR / NFR / BR	Functional Requirement / Non-Functional Requirement / Business Rule
JWT	JSON Web Token used for stateless authentication
JPA	Java Persistence API (used through Hibernate)
LLM	Large Language Model accessed through an external AI provider
MCQ	Multiple-Choice Question
MVP	Minimum Viable Product — the first functional release
RBAC	Role-Based Access Control (roles: Student, Company, Admin)
SPA	Single-Page Application (the React frontend)
SSE	Server-Sent Events — one-way streaming of server messages to the browser over HTTP
LCP	Largest Contentful Paint — a browser metric for how fast the main page content appears
OTP	One-Time Password sent to an email address to prove ownership
CIN / GSTIN	Corporate Identification Number / Goods and Services Tax Identification Number (Indian company registration identifiers)
CTC / LPA	Cost To Company (annual package) / Lakhs Per Annum
Job-match %	Rule-based score showing how well a CV matches a job's requirements (Section 8.2)
Skill gap	Required job skills that are missing or not demonstrated in the CV
Recruiter	A user of the Company Portal acting for a verified company

Term	Meaning
Streak	Number of consecutive calendar days on which a student was active on the platform
Trust level	Label for a company's standing: Verified or, later, Trusted (FR-ADM-14)
Pending_Review	Job status meaning the job waits for an administrator's approval before students can see it

1.4 References

- IEEE Std 830-1998 and ISO/IEC/IEEE 29148 — guidance on SRS structure and requirement quality.
- HireSphere Project Overview, Scope and MVP Plan (companion document, v1.0).
- Official documentation for Spring Boot, Spring Security, React, Vite and MySQL.
- WHATWG HTML Standard (Server-Sent Events) and Web Vitals documentation (Largest Contentful Paint).

1.5 Document Overview

Section 2 gives the overall product description. Section 3 defines external interfaces. Section 4 lists the functional requirements by feature. Section 5 gives detailed use case specifications. Section 6 covers data requirements. Section 7 lists non-functional requirements. Section 8 states business rules and the matching method. The appendices contain a traceability matrix, MVP acceptance criteria, open issues, supporting diagrams and the change log from version 1.0 to 1.1.

Requirement IDs follow the pattern FR-<AREA>-<NN>, NFR-<AREA>-<NN> and BR-<NN>. The word **shall** marks a mandatory requirement.

2. Overall Description

2.1 Product Perspective

HireSphere is a new, self-contained system. It uses a decoupled three-tier architecture: the React SPA runs in the user's browser and communicates with the Spring Boot backend only through secured REST APIs (and a Server-Sent Events stream for long AI responses). The backend stores data in MySQL, keeps CV files and company verification documents in file storage (metadata in MySQL) and calls an external AI provider for language-based features. An email service is needed in the MVP only to verify company email addresses.

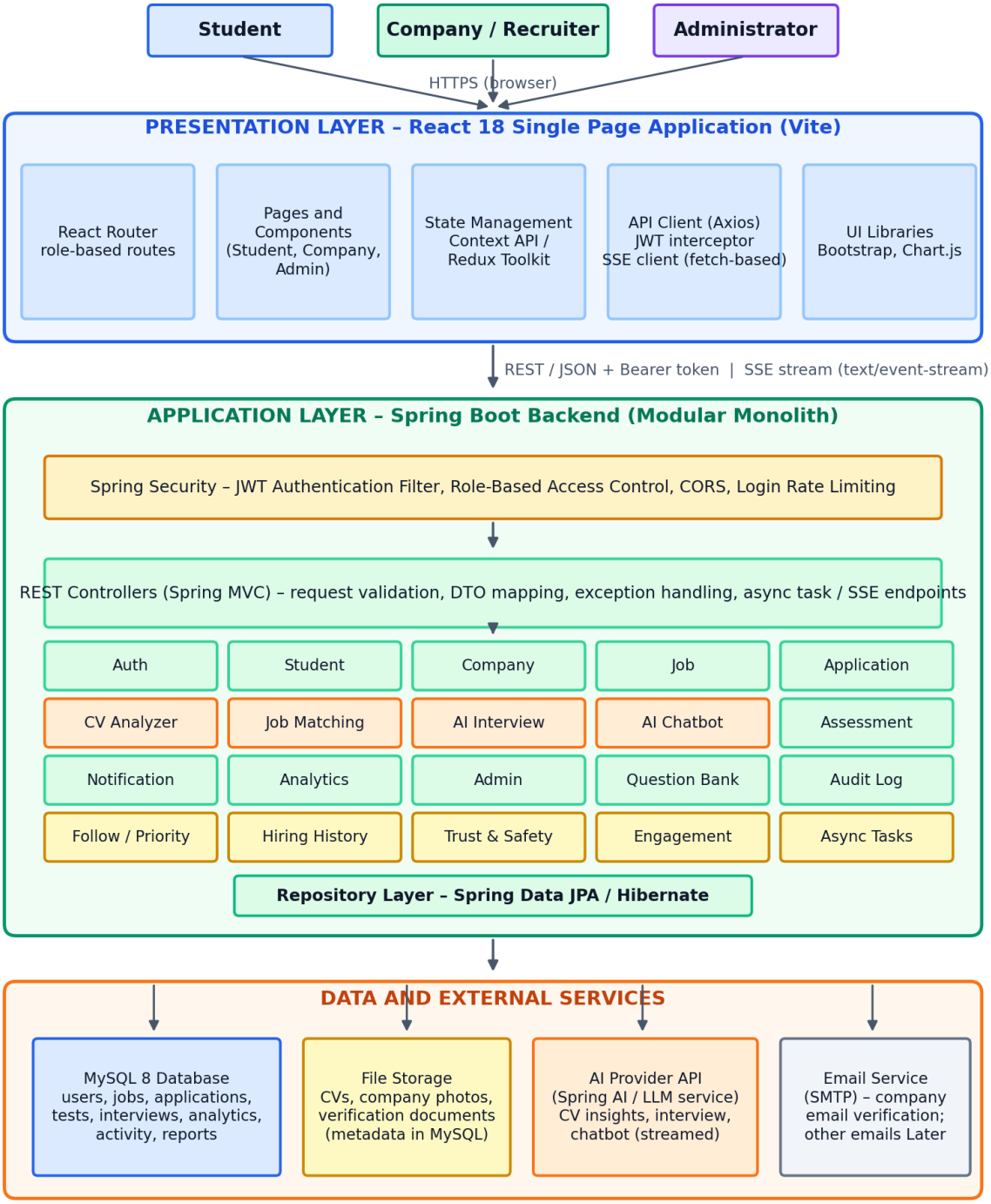


Figure 2.1 — HireSphere system architecture (new components in v1.1 shown in yellow)

2.2 Product Functions (Summary)

- Registration, login and role-based access for students, companies and administrators.
- Job posting, editing, closing and browsing.
- Smart CV Analyzer: CV + job description analysis and CV-only job matching, with ATS check, job-match %, eligibility checklist and skill-gap report.
- Application submission and status tracking.
- Timed MCQ assessments and practice mode with automatic scoring.
- AI mock interviews and an AI career chatbot, with long AI operations run asynchronously and streamed.
- Applicant review, shortlisting and recruitment status management.
- Company profile with platform-verified hiring history (students hired, average and highest package) and career growth paths.
- Following companies with a priority level so that high-priority companies produce priority notifications.
- Student activity streaks and points on the dashboard (Later).
- Fake company and fake job protection: stronger company verification, job screening, student reports and administrator controls.
- Dashboards for students, companies and administrators, and notifications.
- Administration: user management, company verification, job moderation, report handling, audit log and question bank management.

2.3 User Classes and Characteristics

User class	Description	Technical skill	Main needs
Student	Undergraduate or graduate preparing for internships and placements	Basic web use	Improve CV, find suitable jobs, practise, apply and track; trust that companies are real; see company hiring history
Company / Recruiter	HR or hiring staff of a verified organisation	Basic web use	Publish jobs, assess and shortlist candidates, view analytics, present the company and its placement record
Administrator	Platform operator	Moderate	Manage users, verify companies, moderate jobs and reports, maintain question bank, review audit log

2.4 Operating Environment

- Client: modern web browsers (latest two versions of Chrome, Edge, Firefox and Safari) on desktop, tablet and mobile.
- Server: Java 17 or later with Spring Boot, on Windows, Linux or macOS.
- Database: MySQL 8.x.
- External services: an AI provider API reachable over HTTPS, and an email (SMTP) service for company email verification.

2.5 Design and Implementation Constraints

- The frontend shall be built with React and the backend with Java and Spring Boot; MySQL is the primary database.
- All client-server communication shall use REST over HTTPS with JSON payloads and JWT authentication. Streaming AI output uses Server-Sent Events over HTTPS.
- AI features shall use an external AI service; no custom model is trained in the initial version.
- The first release shall be a modular monolith, not a microservice system.
- Uploaded CV text extraction depends on text-based PDFs; scanned image-only CVs may not be analysable.
- The MVP accepts PDF CVs only; DOCX support is deferred (decision recorded as TBD-05).

2.6 Assumptions and Dependencies

- Companies will provide complete job descriptions and eligibility criteria; matching quality depends on this data.
- Students will provide accurate profiles and readable CVs.
- Companies will provide truthful registration details and documents; administrators verify them manually.
- The external AI service is available; when it is not, only AI features are affected.
- Job recommendations and matching initially use only jobs stored in the HireSphere database.
- Hiring history on a company profile reflects only offers recorded through HireSphere unless the company adds approved self-reported data.
- Users have internet access and a supported browser.

2.7 Out of Scope (Initial Version)

- A custom-trained AI model; facial emotion or personality analysis.
- Automatic final hiring decisions without human review.
- Live video proctoring and advanced anti-cheating systems.
- Guaranteed integration with external job portals.
- Automatic verification against government company registries (manual administrator review is used).
- Native Android and iOS apps; enterprise-scale microservices.
- Guaranteed ATS scores or guaranteed placement.

3. External Interface Requirements

3.1 User Interfaces

ID	Requirement	Priority
FR-UI-01	The UI shall be a responsive React application usable from 360 px to 1920 px screen widths.	MVP
FR-UI-02	Each role shall see only its own navigation and pages; unauthorised routes shall redirect to a login or "unauthorised" page.	MVP
FR-UI-03	Forms shall show clear validation messages next to the invalid field.	MVP
FR-UI-04	Long operations (CV analysis, AI responses) shall show a loading indicator or skeleton with progress and a readable error message on failure. Operations that can exceed 5 seconds shall run asynchronously as defined in NFR-PERF-04a to 04f, and the UI shall never block while waiting.	MVP
FR-UI-05	The UI shall support dark and light themes, and the student's choice shall be remembered across sessions. (Merges the former FR-STU-06.)	Later

Principal screens:

Portal	Screens
Public	Landing, Login, Register (student / company), Unauthorised, Not found
Student	Dashboard, CV Analyzer, Job list and details, Company profile and hiring history, Application Tracker, Practice tests and timed test, AI mock interview, Career chatbot, Followed companies, Report job, Profile and settings, Activity and points (Later)
Company	Dashboard, Manage jobs, Job form, Applicants and shortlisting, Create tests and results, Company profile and gallery, Hiring history and career paths, Verification documents
Admin	Dashboard, Users, Company verification (with checklist), Job moderation and review queue, Reports queue, Question bank, Announcements, Audit log

3.2 Hardware Interfaces

HireSphere has no direct hardware interfaces. Users need a device with a web browser; microphone and camera are not required in the initial version (voice and video interviews are deferred).

3.3 Software Interfaces

Interface	Purpose	Details
MySQL 8.x	Persistent storage	Accessed via Spring Data JPA / Hibernate over JDBC
AI provider API	CV insights, interview questions and feedback, chatbot	HTTPS requests with prompts; API key kept in environment configuration; accessed via Spring AI or a provider SDK; responses can be streamed
File storage	CV files, company images and company verification documents	Local or cloud storage outside the public web root; MySQL keeps metadata and storage path
Document parsing library	CV text extraction	Apache PDFBox or Apache Tika
Email service (SMTP)	Company email verification link / OTP in the MVP; other email notifications Later	Credentials kept in environment configuration; provider to be chosen (TBD-03)

3.4 Communication Interfaces

- REST APIs over HTTPS exchanging JSON; file uploads use multipart/form-data.
- Authentication via an Authorization: Bearer <JWT> header on every protected request.
- Long AI operations return 202 Accepted with a task ID, or open a Server-Sent Events (text/event-stream) response; see NFR-PERF-04.
- CORS shall allow only the configured frontend origin(s).

4. System Features and Functional Requirements

Each feature below gives a short description and its functional requirements. Priority MVP means the requirement belongs to the first release; Later means it is planned for a subsequent release. Rows shaded yellow are new or changed in v1.1.

4.1 Authentication and User Management

Users register, log in and are restricted to the functions of their role.

ID	Requirement	Priority
FR-AUTH-01	The system shall allow a student to register with name, email, password and basic academic details.	MVP
FR-AUTH-02	The system shall allow a company to register with company name, contact person, official work email, password, contact details, registration number, website and a proof document (see FR-ADM-08); the account shall remain Pending until verified by an administrator.	MVP
FR-AUTH-03	The system shall authenticate users by email and password and return a JWT on success.	MVP
FR-AUTH-04	The system shall store passwords only as salted hashes (BCrypt) and shall never return them in any response.	MVP
FR-AUTH-05	The system shall enforce role-based access (STUDENT, COMPANY, ADMIN) on every backend endpoint and on every frontend route.	MVP
FR-AUTH-06	The system shall reject duplicate email registrations and invalid input with a clear message.	MVP
FR-AUTH-07	The system shall treat an expired or invalid token as unauthenticated and redirect the user to the login page.	MVP
FR-AUTH-08	An administrator shall be able to activate or deactivate any account; a deactivated user shall not be able to log in.	MVP
FR-AUTH-09	The system shall support password reset through an email link.	Later

4.2 Student Profile

Students maintain the academic and professional information used for eligibility checks and matching.

ID	Requirement	Priority
FR-STU-01	The student shall be able to view and edit personal details, college, degree, graduation year and CGPA.	MVP
FR-STU-02	The student shall be able to add, edit and remove skills, projects and certifications.	MVP
FR-STU-03	The student shall be able to set preferred job roles and locations.	MVP
FR-STU-04	The student shall be able to upload, replace and delete a CV (PDF only in the MVP; DOCX is Later), maximum 5 MB.	MVP
FR-STU-05	The student shall be able to set notification preferences.	Later
FR-STU-06	Withdrawn: merged into FR-UI-05 (dark / light mode). The ID is kept so earlier references stay traceable.	—

4.3 Job Management and Browsing

Verified companies publish and manage vacancies; students browse them. Job statuses are defined in Section 8.4.

ID	Requirement	Priority
FR-JOB-01	A verified company shall be able to create a job with title, description, required skills (at least one), eligibility criteria (for example minimum CGPA, degree, graduation year), salary or range, and application deadline.	MVP

ID	Requirement	Priority
FR-JOB-02	The company shall be able to add optional details: preferred skills, experience, working hours and days, leave policy, benefits, work-from-home or hybrid option, and selection rounds.	Later
FR-JOB-03	The company shall be able to edit and close its own vacancies.	MVP
FR-JOB-04	The system shall automatically stop accepting applications when the deadline passes or the job is closed.	MVP
FR-JOB-05	The system shall prevent companies whose verification status is not Verified (Pending, Rejected or Suspended) from publishing jobs.	MVP
FR-JOB-06	Students shall be able to browse published jobs and filter by company, location, salary and work mode.	MVP
FR-JOB-07	Students shall be able to view full job details including eligibility, deadline and the company's Verified badge.	MVP
FR-JOB-08	An administrator shall be able to flag, hide or remove invalid or reported job postings (statuses Pending_Review, Hidden and Removed).	MVP
FR-JOB-09	The first 3 jobs of every newly verified company shall be held as Pending_Review until an administrator approves them.	MVP
FR-JOB-10	The system shall scan job text for scam indicators (initial list in Section 8.6), for example registration fee, security deposit, pay to apply, WhatsApp or Telegram-only contact, guaranteed job and personal email. Flagged jobs shall be held as Pending_Review and placed in the administrator review queue.	MVP
FR-JOB-11	Students shall be able to report a job or company with a reason. When reports on a job reach a threshold (default 3 from different students, configurable), the job shall be hidden automatically until an administrator reviews it.	MVP
FR-JOB-12	The system shall limit the number of job postings per company per day (configurable).	Later
FR-JOB-13	Jobs and company profiles shall show a Verified badge, and job pages shall show the warning "Never pay money to apply for a job. Report suspicious jobs."	MVP

4.4 Smart CV Analyzer

The analyzer compares a student's CV with a job description (Case 1) or with all published jobs (Case 2) and produces a structured report. ATS compatibility, job-match percentage and eligibility are reported separately.

ID	Requirement	Priority
FR-CV-01	The system shall validate an uploaded CV for allowed file type (PDF in the MVP), size and readable content before analysis.	MVP
FR-CV-02	The system shall extract text from the CV and identify skills, education, projects, certifications and experience.	MVP
FR-CV-03	Case 1: the system shall accept a job description (pasted text or file) or an existing vacancy selected by the student.	MVP
FR-CV-04	Case 2: the system shall compare the CV with all published jobs and show a separate match result for each.	MVP
FR-CV-05	The system shall calculate a job-match percentage using the weighted method in Section 8.2.	MVP
FR-CV-06	The report shall list matching skills and missing skills (skill gap).	MVP
FR-CV-07	The report shall include an eligibility checklist; where required student data is unavailable it shall show "cannot be determined" instead of passing or failing.	MVP
FR-CV-08	The report shall include an indicative ATS compatibility result based on readability, structure and keyword coverage.	MVP
FR-CV-09	The system shall generate suggestions for skills, projects, certifications and CV improvements using the AI service. This call shall run asynchronously (NFR-PERF-04) and the rule-based results shall be shown first.	MVP
FR-CV-10	In Case 2, results shall be ordered by match percentage, with links to job details and the Apply action.	MVP

ID	Requirement	Priority
FR-CV-11	The system shall store each analysis result and skill-gap report so the student can view past reports.	MVP
FR-CV-12	Every report shall state that results are indicative and do not guarantee any ATS result, eligibility or selection.	MVP
FR-CV-13	If text cannot be extracted (for example a scanned image CV), the system shall tell the student and suggest uploading a text-based PDF.	MVP
FR-CV-14	If the AI service is unavailable, the system shall still show the rule-based match, skills and eligibility results and indicate that AI suggestions are temporarily unavailable.	MVP
FR-CV-15	The system shall show a consent prompt before the student's first use of any AI feature and store the consent with its date. Without consent, only rule-based results are shown and no CV or interview content is sent to the AI service (implements NFR-PRV-01).	MVP

4.5 Job Recommendations and Alerts

Students are told about relevant opportunities.

ID	Requirement	Priority
FR-REC-01	The system shall notify students when a newly published job matches their preferred roles or skills (basic rule).	MVP
FR-REC-02	The system shall show a list of recommended jobs on the student dashboard based on skills, education and preferred roles.	MVP
FR-REC-03	The system shall support advanced alert rules and configurable alert preferences.	Later

4.6 Application Tracker

Students apply to jobs and follow progress; companies update progress.

ID	Requirement	Priority
FR-APP-01	A student shall be able to apply once to an open job before its deadline. An application cannot be resubmitted after withdrawal (BR-02).	MVP
FR-APP-02	The system shall record the submission date and set the status to Applied.	MVP
FR-APP-03	The student shall see a list of applications with company, job, submission date and current status.	MVP
FR-APP-04	Allowed statuses shall be Applied, Assessment, Shortlisted, Interview, Selected, Rejected and Withdrawn, following the transitions in Section 8.1.	MVP
FR-APP-05	Every status change shall be saved in a status history with time and the user who made it.	MVP
FR-APP-06	A student shall be able to withdraw an application until a final decision is recorded. Withdrawal is final.	MVP
FR-APP-07	The system shall create a notification for the student on every status change.	MVP
FR-APP-08	If the student does not meet explicit eligibility criteria, the system shall warn before applying; blocking is configurable per job (default: warn only).	MVP
FR-APP-09	The tracker shall show assessment and interview schedules and upcoming deadlines.	Later

4.7 Assessments and Practice

Companies assess candidates with timed MCQ tests; students practise.

ID	Requirement	Priority
FR-TST-01	A company shall be able to create a test for a job by selecting questions from the question bank or adding its own, and setting a duration.	MVP
FR-TST-02	The company shall be able to assign a test to selected applicants.	MVP
FR-TST-03	A student shall be able to attempt an assigned test within the time limit; the attempt shall auto-submit when time ends.	MVP

ID	Requirement	Priority
FR-TST-04	The system shall automatically evaluate MCQ answers and store score and answers in MySQL.	MVP
FR-TST-05	A student shall have one attempt per assigned test unless the company allows a retake.	MVP
FR-TST-06	The company shall see results for each candidate and be able to sort by score.	MVP
FR-TST-07	Practice mode shall let students attempt aptitude and technical MCQs by topic, company and difficulty and see correct answers with explanations.	MVP
FR-TST-08	The system shall keep a practice performance history and show comparison with aggregate results.	Later
FR-TST-09	The system shall support coding problems with execution in a secure sandbox.	Later
FR-TST-10	The system shall support advanced online proctoring.	Later

4.8 AI Interview Agent

Students practise interviews; companies later run preliminary AI interviews.

ID	Requirement	Priority
FR-INT-01	A student shall be able to start a text-based mock interview by choosing type (HR or technical) and a target role.	MVP
FR-INT-02	The system shall generate interview questions relevant to the chosen role and the student's skills.	MVP
FR-INT-03	The student shall submit answers as text.	MVP
FR-INT-04	The system shall return AI feedback on relevance, completeness and clarity with improvement suggestions. Feedback shall be streamed to the UI (NFR-PERF-04b).	MVP
FR-INT-05	At the end of a session the system shall present a summary report and keep the session in the student's history.	MVP
FR-INT-06	If the AI service fails or times out, the system shall send an error event, show an error, keep answers and any partial output already received, and allow retry.	MVP
FR-INT-07	A company shall be able to configure a job-specific preliminary interview with question sets and an evaluation rubric.	Later
FR-INT-08	The system shall produce an AI assessment summary for recruiter review, clearly labelled as a recommendation.	Later
FR-INT-09	The system shall support voice and video interviews.	Later

4.9 AI Career Chatbot

A conversational guide for career questions.

ID	Requirement	Priority
FR-BOT-01	The system shall provide a chat interface where a student can ask career, CV, interview, skill and job-search questions. Answers shall be streamed (NFR-PERF-04b).	MVP
FR-BOT-02	The system shall store chat history per user and show it on return.	MVP
FR-BOT-03	The system shall display a notice that responses are AI-generated guidance.	MVP
FR-BOT-04	If the AI service is unavailable or times out, the chatbot shall show a clear message, keep any partial answer and allow retry.	MVP
FR-BOT-05	With the student's consent, the chatbot shall use profile and CV information for personalised answers.	Later

4.10 Applicant Management and Shortlisting

Companies review applicants and manage recruitment.

ID	Requirement	Priority
FR-CAM-01	A company shall see all applicants for its own jobs with profile, CV, eligibility information and CV match result.	MVP

ID	Requirement	Priority
FR-CAM-02	The company shall be able to filter and sort applicants by status, test score and eligibility.	MVP
FR-CAM-03	The company shall be able to change an applicant's status and shortlist candidates. Setting the status to Selected requires the offered role and package (FR-COM-05).	MVP
FR-CAM-04	The company shall be able to view test results and, later, AI interview feedback.	MVP
FR-CAM-05	A company shall not be able to see applicants or data of another company's jobs.	MVP
FR-CAM-06	The system shall not automatically select or reject any candidate; every final status change shall be made by a company user.	MVP

4.11 Company Profile

Companies present themselves to students, including their hiring track record.

ID	Requirement	Priority
FR-COM-01	A company shall be able to edit its description, industry, contact details and work arrangements.	MVP
FR-COM-02	Students shall be able to view a company's profile from a job page, including its Verified badge and aggregated hiring history.	MVP
FR-COM-03	A company shall be able to upload and manage workplace photographs and event images (image types only, size-limited).	Later
FR-COM-04	The company profile shall show hiring history per academic year: number of students hired, average package, highest package and roles offered.	MVP
FR-COM-05	When a recruiter sets an application to Selected, the system shall require the offered role and package (CTC in LPA). Hiring history shall be computed automatically from these records and labelled "Platform-verified".	MVP
FR-COM-06	A company may add past hiring data from before it joined HireSphere. It shall be labelled "Self-reported" and shall be visible only after administrator approval.	Later
FR-COM-07	Students shall see only aggregated figures. Individual offers shall never be shown, and averages and maxima shall be hidden when fewer than 3 hires exist, to protect privacy (BR-14).	MVP
FR-COM-08	A company shall be able to describe career growth paths per role (for example Trainee → Engineer → Senior Engineer) with the typical time per level and growth in package. They are shown on the profile after administrator approval.	Later

4.12 Dashboards and Analytics

Role-specific overviews.

ID	Requirement	Priority
FR-DSH-01	The student dashboard shall show available jobs, application statuses, CV analysis summaries and test results.	MVP
FR-DSH-02	The company dashboard shall show published jobs, applicant counts, assessment results and shortlisted candidates.	MVP
FR-DSH-03	The admin dashboard shall show total students, companies, active jobs and applications, the number of pending company verifications and open job reports, with links to management pages.	MVP
FR-DSH-04	The Career Readiness Dashboard shall combine practice results, CV analysis and mock interview history and suggest areas for improvement.	Later
FR-DSH-05	The Recruitment Analytics Dashboard shall show applicants per role, stage conversion rates, time-to-hire and graphical reports.	Later
FR-DSH-06	The system shall record one activity day per student per calendar day.	Later
FR-DSH-07	The student dashboard shall show total active days, current streak, longest streak (one grace day per week keeps a streak alive) and a monthly calendar view.	Later
FR-DSH-08	The system shall award points for meaningful activities according to Section 8.5, with a daily cap (default 50, configurable).	Later

ID	Requirement	Priority
FR-DSH-09	The dashboard shall show total points, level or badge and a weekly points trend. Points are for motivation only (BR-12).	Later

4.13 Notifications

Users are informed of relevant events. Students can also decide which companies matter most to them.

ID	Requirement	Priority
FR-NOT-01	The system shall create in-app notifications for status changes, test assignments, company verification outcome, job review outcome (approved or removed), company suspension and new matching jobs.	MVP
FR-NOT-02	Users shall be able to view notifications and mark them as read.	MVP
FR-NOT-03	The system shall send email notifications for key events.	Later
FR-NOT-04	A student shall be able to follow companies and set a priority for each: High, Normal or Muted.	MVP
FR-NOT-05	When a followed company publishes a job (status Open), the student shall get a notification. For High priority the notification shall be marked "Priority" and listed first. Muted companies produce no notification.	MVP
FR-NOT-06	For High-priority companies, the system shall also notify the student when a job deadline is within 48 hours.	MVP
FR-NOT-07	Jobs from followed High-priority companies shall be highlighted in the job list and shown earlier in recommendations. This shall not change the match %.	Later
FR-NOT-08	The number of High-priority companies per student shall be limited (default 10, configurable) to avoid notification overload.	MVP
FR-NOT-09	High-priority alerts shall also be sent by email.	Later
FR-NOT-10	A student shall not receive two notifications for the same job; when a job matches both FR-REC-01 and a followed company, only the higher-priority notification is kept.	MVP

4.14 Administration

Platform operators maintain users, companies, jobs and content, and protect students from fake companies and fake jobs. No automated check can fully prove that a company is real, so protection is layered and ends with human review.

A. Users, content and audit

ID	Requirement	Priority
FR-ADM-01	An administrator shall be able to list, search, activate and deactivate student and company accounts.	MVP
FR-ADM-02	An administrator shall be able to review company registrations and approve or reject them with a reason, after completing the checklist in FR-ADM-10.	MVP
FR-ADM-03	An administrator shall be able to monitor and moderate job postings, including the Pending_Review queue and flagged jobs.	MVP
FR-ADM-04	An administrator shall be able to add, edit and delete aptitude and technical MCQs, organised by topic, company and difficulty.	MVP
FR-ADM-05	Administrator accounts shall not be created through public registration.	MVP
FR-ADM-06	An administrator shall be able to manage announcements and FAQs.	Later
FR-ADM-07	The system shall keep an append-only audit log of administrative actions and allow administrators to review it.	MVP

B. Stronger company verification

ID	Requirement	Priority
FR-ADM-08	Company registration shall also require an official registration number (for example CIN or GSTIN), company website, official work email and at least one proof document (PDF or image, size-limited).	MVP

ID	Requirement	Priority
FR-ADM-09	The system shall verify email ownership by link or OTP. Free-mail domains (Gmail, Yahoo and similar) shall be flagged for extra review.	MVP
FR-ADM-10	The administrator shall complete a checklist before approval: registration number checked, email domain matches website, website reachable, document reviewed, and no duplicate registration number, domain or phone on another account. The decision and reason shall be logged.	MVP
FR-ADM-11	The system shall detect duplicate registration numbers, domains and phone numbers across accounts and warn the administrator.	MVP

C. Controls and trust

ID	Requirement	Priority
FR-ADM-12	An administrator shall be able to suspend a company or revoke its verification. All its jobs shall be hidden and affected applicants notified.	MVP
FR-ADM-13	Every verification, rejection, suspension, report decision and moderation action shall be written to the audit log of FR-ADM-07 with administrator, time, target, decision and reason.	MVP
FR-ADM-14	Companies shall have a trust level: Verified, then Trusted after a configurable number of jobs with no upheld reports.	Later
FR-ADM-15	An administrator shall be able to view the reports queue, see report history per job and company, and uphold or dismiss each report with a reason.	MVP

5. Use Case Specifications

The use case diagrams show the actors and goals of each portal. Detailed specifications of the key use cases follow. Each diagram is drawn at full width so that all labels are readable.

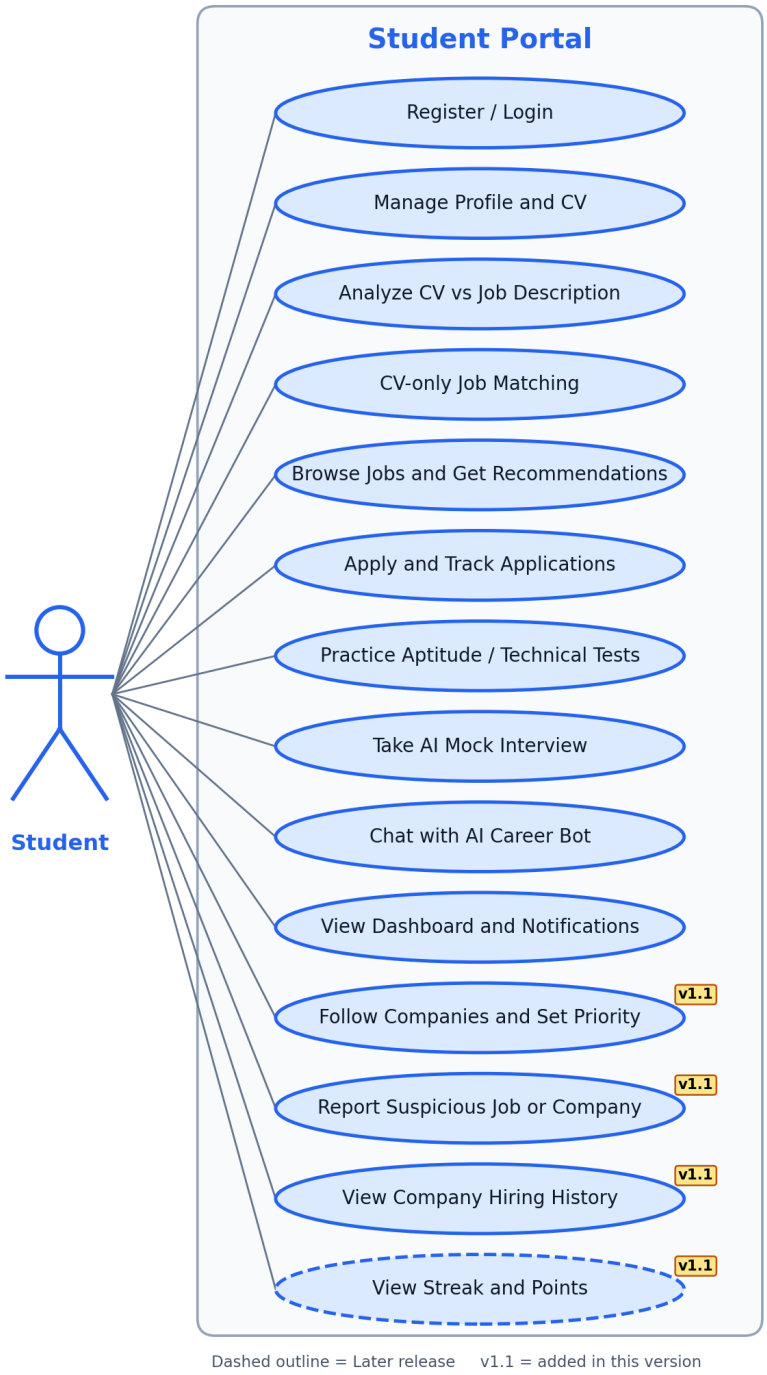


Figure 5.1 — Student Portal use cases

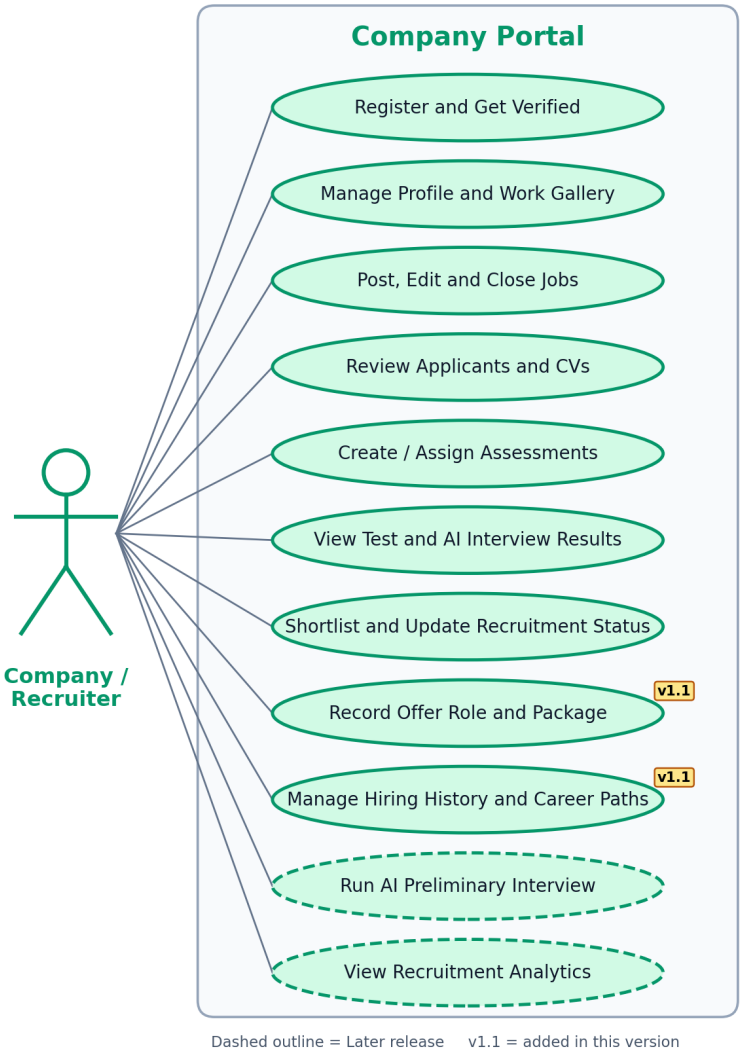


Figure 5.2 — Company Portal use cases

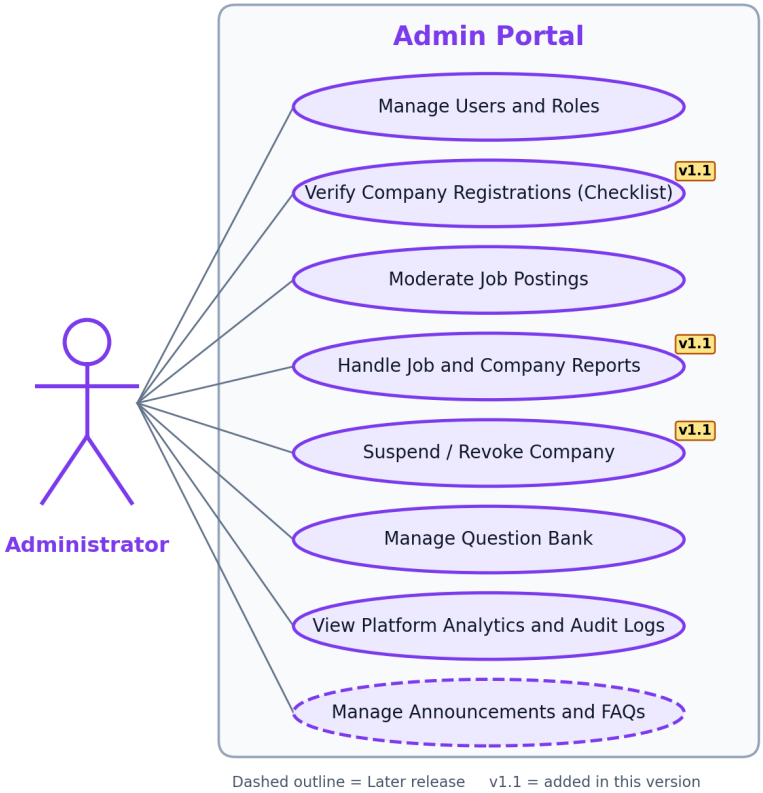


Figure 5.3 — Admin Portal use cases

UC-01 Register and Log In

Actor	Student, Company
Description	A new user creates an account and later logs in to the portal matching their role.
Preconditions	The user has a valid email address and is not logged in.
Main flow	1. The user opens the registration page and selects Student or Company. 2. The user enters the required details and submits. 3. The system validates the input and creates the account (companies are set to Pending). 4. The user opens the login page and enters email and password. 5. The system verifies the credentials and returns a JWT. 6. The system redirects the user to the dashboard of their role.
Alternate / exception flows	

Ref	Step	Condition	System response
2a	2	Email already registered or input invalid.	Shows an error next to the field; no account is created.
2b	2	Company registration lacks registration number, website, work email or proof document.	Shows a validation error; no account is created.
3a	3	A company account is created.	Account is set to Pending; email verification link or OTP is sent (FR-ADM-09).
5a	5	Wrong credentials or deactivated account.	Shows a generic error; the user is not logged in.
5b	5	Too many failed logins (NFR-SEC-09).	Blocks further attempts temporarily and shows a wait message.
6a	6	Company is still Pending.	Shows the dashboard with recruitment features disabled and a verification notice.
6b	6	Company is Rejected or Suspended.	Shows the dashboard with the status and reason; posting stays disabled.

Postconditions	The user has an authenticated session limited to their role.
Related requirements	FR-AUTH-01 to FR-AUTH-08, FR-ADM-08, FR-ADM-09, NFR-SEC-09

UC-02 Verify Company Registration

Actor	Administrator
Description	The administrator reviews a pending company, completes the verification checklist and approves or rejects it.
Preconditions	The administrator is logged in and at least one company is Pending.
Main flow	1. The administrator opens the company verification page. 2. The system lists pending companies, marking free-mail domains and possible duplicates. 3. The administrator opens a company and reviews its details and proof documents. 4. The administrator completes the checklist: registration number checked, email domain matches website, website reachable, document reviewed, no duplicate registration number, domain or phone. 5. The administrator selects Approve. 6. The system sets the company to Verified, writes the audit log and notifies the company.
Alternate / exception flows	4a. The checklist is incomplete: the Approve action stays disabled. 5a. The administrator selects Reject and enters a reason; the system sets the company to Rejected, logs the decision and notifies it.
Postconditions	Verified companies can publish jobs; rejected companies cannot.
Related requirements	FR-ADM-02, FR-ADM-08 to FR-ADM-11, FR-ADM-13, FR-JOB-05, FR-NOT-01

UC-03 Post Job Vacancy	
Actor	Company (verified)
Description	A recruiter publishes a vacancy with its requirements and eligibility criteria.
Preconditions	The company is logged in and Verified.
Main flow	1. The recruiter opens Post Job. 2. The recruiter enters title, description, required skills, eligibility criteria, salary and deadline. 3. The system validates the data (deadline in the future, required fields present). 4. The recruiter publishes the job. 5. The system screens the text for scam indicators and checks whether one of the company's first 3 jobs applies. If clean, it saves the job as Open and makes it visible to students. 6. The system notifies students whose preferences match the job and students who follow the company.
Alternate / exception flows	3a. Validation fails: the system highlights the fields and does not save. 4a. The recruiter saves as draft; the job is not visible to students. 5a. The job is flagged or is one of the first 3 jobs: it is saved as Pending_Review and the recruiter sees "Awaiting admin review". When approved the job becomes Open and step 6 follows; if removed the company is notified with the reason. After publishing: the recruiter can edit or close the job.
Postconditions	The job is stored and visible in the job list until its deadline or closure (once Open).
Related requirements	FR-JOB-01 to FR-JOB-05, FR-JOB-09, FR-JOB-10, FR-REC-01, FR-NOT-05

UC-04 Analyze CV against a Job Description (Case 1)	
Actor	Student
Description	The student checks how well their CV matches one specific job.
Preconditions	The student is logged in and has uploaded a CV or uploads one now.
Main flow	1. The student opens the CV Analyzer and chooses "CV + Job Description". 2. The student uploads or selects a CV (PDF). 3. The student pastes or uploads a job description, or selects an existing vacancy. 4. The system validates the file and extracts the CV text. 5. The system identifies skills and other CV information and extracts the job requirements. 6. The system calculates job-match %, checks eligibility and the ATS indicators and shows these rule-based results immediately. 7. If the student has given AI consent, the system requests improvement suggestions from the AI service asynchronously; they are streamed or loaded into the report when ready. 8. The system stores the report: ATS result, match %, eligibility checklist, matching skills, missing skills and recommendations.
Alternate / exception flows	4a. Invalid type or size: the system rejects the file with a message. 4b. No text extractable: the system advises uploading a text-based PDF. 6a. No AI consent yet: the system asks for consent (FR-CV-15); if declined, only the rule-based report is shown. 7a. AI unavailable or 30-second limit reached: the report is shown without AI suggestions, with a notice and a Retry option.
Postconditions	The report is saved and appears in the student's analysis history.
Related requirements	FR-CV-01 to FR-CV-03, FR-CV-05 to FR-CV-09, FR-CV-11 to FR-CV-15, NFR-PERF-04

UC-05 Match CV against Available Jobs (Case 2)	
Actor	Student
Description	The student uploads only a CV and sees how it matches the jobs on the platform.
Preconditions	The student is logged in; at least one job is published.
Main flow	1. The student chooses "CV only" in the CV Analyzer and uploads or selects a CV. 2. The system extracts the CV text and skills. 3. The system compares the CV with every open job. 4. The system displays a list ordered by match %, each with matching skills, missing skills and eligibility result. 5. The student opens a job for details or selects Apply.
Alternate / exception flows	3a. No jobs are published: the system shows a message that no vacancies are available.
Postconditions	Match results are stored; the student may apply to any listed job.
Related requirements	FR-CV-01, FR-CV-02, FR-CV-04 to FR-CV-07, FR-CV-10 to FR-CV-15

UC-06 Apply for Job and Track Application	
Actor	Student
Description	The student applies to a job and follows its status.
Preconditions	The student is logged in; the job is Open and before its deadline.
Main flow	1. The student opens a job and selects Apply. 2. The system checks the deadline, duplicate application and eligibility. 3. The system records the application with status Applied and the submission date. 4. The student opens the Application Tracker and sees the application. 5. When the company changes the status, the system records it in history and notifies the student.
Alternate / exception flows	2a. Already applied (including a withdrawn application) or deadline passed: the system blocks the action with a message. 2b. Eligibility criteria not met: the system warns; the student may continue unless the job blocks it. 5a. The student withdraws before a final decision; status becomes Withdrawn and cannot be reversed.
Postconditions	The application and its status history are stored and visible to the student and the owning company.
Related requirements	FR-APP-01 to FR-APP-08, FR-NOT-01, BR-02

UC-07 Review Applicants and Update Status	
Actor	Company
Description	The recruiter reviews applicants of a job, uses assessment results, and shortlists candidates.
Preconditions	The recruiter is logged in; the job belongs to the company and has applicants.
Main flow	1. The recruiter opens the applicants list of a job. 2. The system shows applicants with profile, CV, eligibility and match information. 3. The recruiter filters or sorts by status, test score or eligibility. 4. The recruiter opens a candidate and reviews details and test results. 5. The recruiter changes the status (for example Shortlisted or Rejected). For Selected the system asks for the offered role and package (LPA). 6. The system records the change, notifies the student and, for Selected, updates the company's hiring history.
Alternate / exception flows	5a. The requested transition is not allowed by the lifecycle: the system rejects it with a message. 5b. Selected without role or package: the system rejects the change and asks for the missing values. Access to another company's job is denied.
Postconditions	The status and history are updated; final decisions are made only by the recruiter.
Related requirements	FR-CAM-01 to FR-CAM-06, FR-APP-04, FR-APP-05, FR-COM-04, FR-COM-05, FR-COM-07

UC-08 Attempt Timed Test	
Actor	Student
Description	The student completes an assigned MCQ test within the time limit.
Preconditions	The student has an assigned test that has not been attempted.
Main flow	1. The student opens the assigned test and starts the attempt. 2. The system starts the timer and shows the questions. 3. The student selects answers and submits. 4. The system evaluates the answers, stores the score and answers. 5. The company can view the result; the student sees a confirmation.
Alternate / exception flows	1a. The attempt was already made: the system blocks a second attempt unless a retake is allowed. 3a. Time expires: the system auto-submits the answers given so far. 3b. Connection lost: answers saved so far are kept and the timer continues on the server.
Postconditions	The attempt, answers and score are stored in MySQL.
Related requirements	FR-TST-02 to FR-TST-06

UC-09 Take AI Mock Interview	
Actor	Student
Description	The student practises an interview and receives AI feedback.
Preconditions	The student is logged in and has given AI consent.
Main flow	1. The student selects interview type (HR or technical) and a target role. 2. The system generates the first question using the AI service. 3. The student types an answer and submits. 4. The system streams AI feedback on relevance, completeness and clarity, with suggestions. 5. Steps 2–4 repeat for the configured number of questions. 6. The system shows a summary report and saves the session.
Alternate / exception flows	2a/4a. AI service fails or times out: the system sends an error, keeps earlier answers and partial output, and allows retry.
Postconditions	The session, questions, answers and feedback are saved in the student's history.
Related requirements	FR-INT-01 to FR-INT-06, FR-CV-15

UC-10 Chat with AI Career Bot	
Actor	Student
Description	The student asks career-related questions and receives guidance.
Preconditions	The student is logged in and has given AI consent.
Main flow	1. The student opens the chatbot and types a question. 2. The system sends the question with recent history to the AI service. 3. The system streams the answer to the screen and saves the exchange.
Alternate / exception flows	3a. AI unavailable or timed out: the system shows an error, keeps any partial answer and allows retry.
Postconditions	The conversation is stored in the chat history.
Related requirements	FR-BOT-01 to FR-BOT-04

UC-11 Follow Company and Receive Priority Alerts	
Actor	Student
Description	The student follows companies and marks the ones that matter most so that their jobs produce priority notifications.
Preconditions	The student is logged in.
Main flow	1. The student opens a company profile or the Followed companies page and selects Follow. 2. The student sets the priority: High, Normal or Muted. 3. The system stores the follow and its priority. 4. When the company publishes a job (status Open), the system creates a notification; for High priority it is marked Priority and listed first. 5. For High priority, the system sends a further notification when a job deadline is within 48 hours.
Alternate / exception flows	2a. The student already has the maximum number of High-priority companies (default 10): the system asks them to lower another one first. 4a. Priority is Muted: no notification is created. 4b. The job also matches the student's preferred roles or skills: only one notification, the higher-priority one, is created.
Postconditions	The follow and priority are stored; notifications follow the chosen priority.
Related requirements	FR-NOT-04 to FR-NOT-10, FR-REC-01

UC-12 Report Suspicious Job or Company	
Actor	Student
Description	The student reports a job or company that looks fake or asks for money.
Preconditions	The student is logged in and viewing a job or company page.
Main flow	1. The student selects Report on the job or company page. 2. The student chooses a reason (for example asks for payment, fake company, misleading details) and may add a comment. 3. The system stores the report and thanks the student. 4. When reports on a job reach the threshold (default 3 from different students), the system hides the job, adds it to the reports queue and shows it on the admin dashboard.
Alternate / exception flows	3a. The student has already reported this job: the system shows a message and stores nothing new.
Postconditions	The report is stored; the job may be hidden pending administrator review.
Related requirements	FR-JOB-11, FR-JOB-13, FR-ADM-15

UC-13 Handle Reports and Suspend Company	
Actor	Administrator
Description	The administrator reviews reported or flagged jobs and companies and takes action.
Preconditions	The administrator is logged in and at least one report or Pending_Review job exists.
Main flow	1. The administrator opens the reports queue. 2. The system lists items with job, company, report count and report history. 3. The administrator reviews the job, the company profile and its documents. 4. The administrator upholds the report (job set to Removed) or dismisses it (job restored to Open). 5. If the company is fake, the administrator suspends it or revokes its verification; all its jobs are hidden and affected applicants are notified. 6. The system writes every decision with its reason to the audit log.
Alternate / exception flows	4a. A flagged job in Pending_Review is approved: it becomes Open and matching students and followers are notified.
Postconditions	Job and company statuses are updated and the audit log is complete.
Related requirements	FR-ADM-12, FR-ADM-13, FR-ADM-15, FR-JOB-08, FR-JOB-09, FR-JOB-10, FR-NOT-01

6. Data Requirements

6.1 Logical Data Model

The ER diagram is split into three figures so that every table stays readable. Supporting tables (resume skills, test answers, recruitment rounds, shortlists, announcements, learning progress and others) follow the same foreign-key pattern. Yellow cells are new or changed in v1.1; dashed grey boxes repeat a table from Figure 6.1a.

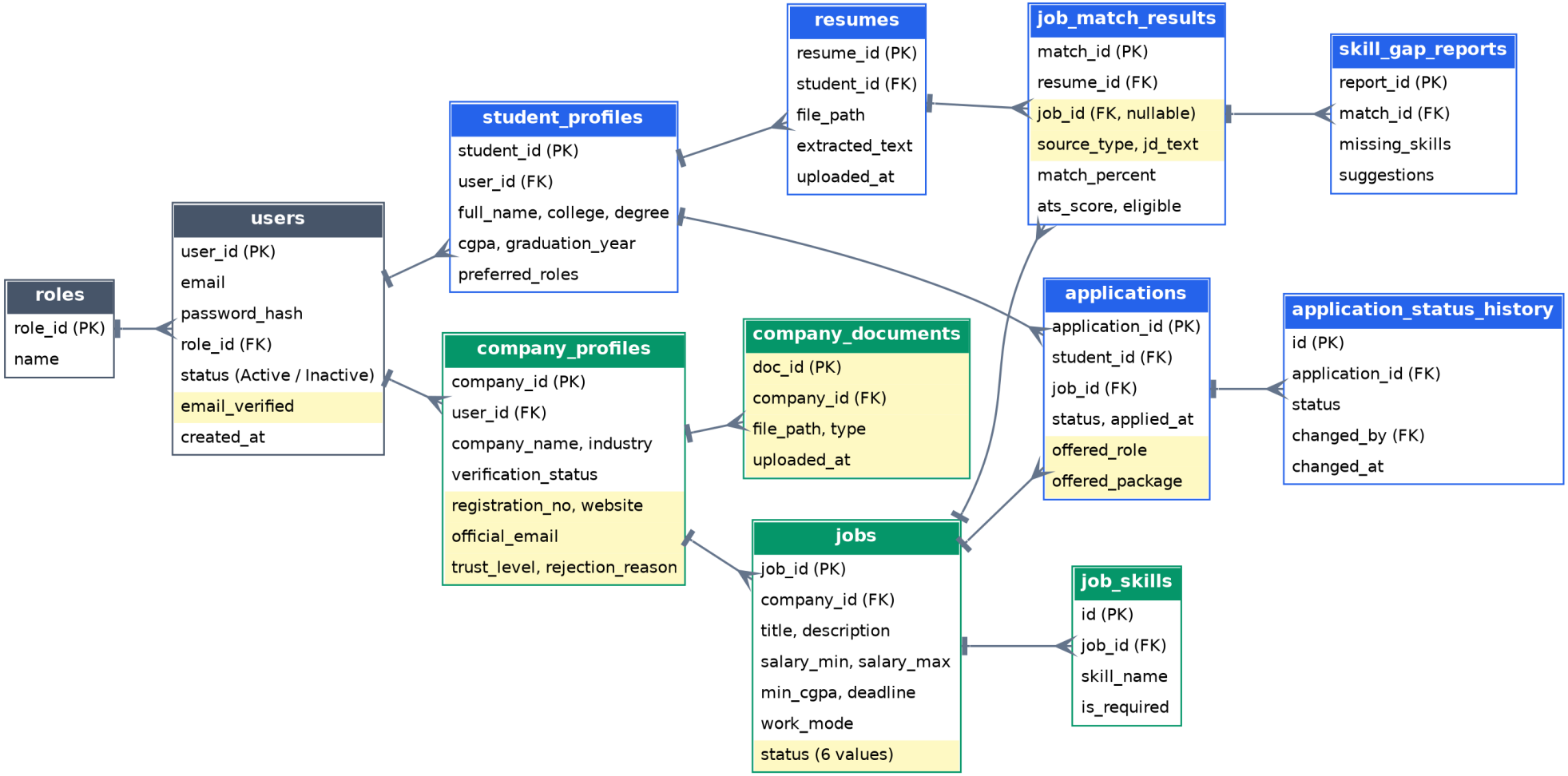


Figure 6.1a — Core entities: accounts, profiles, CV matching, jobs and applications

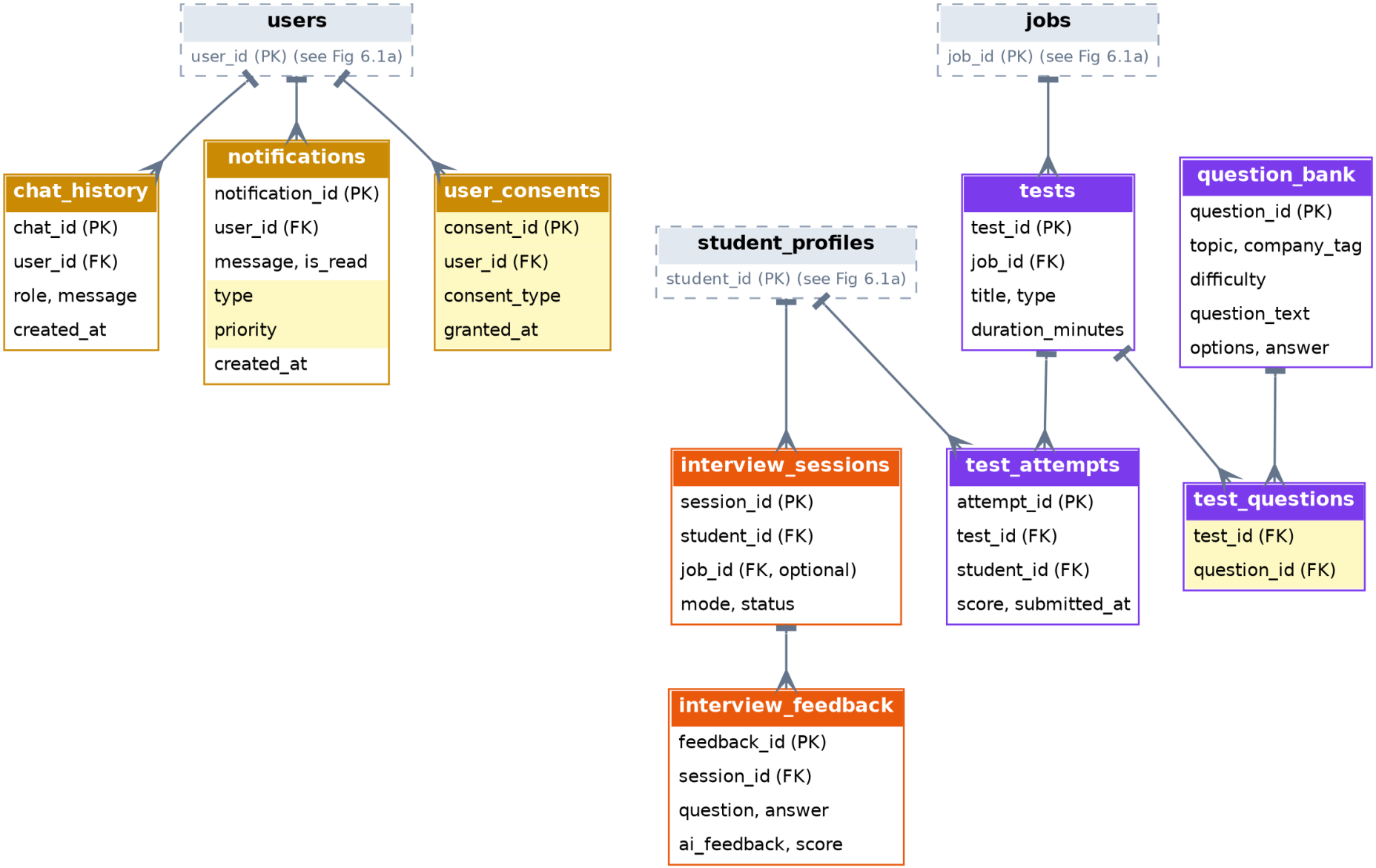


Figure 6.1b — Assessments, AI interview, chat, notifications and consent

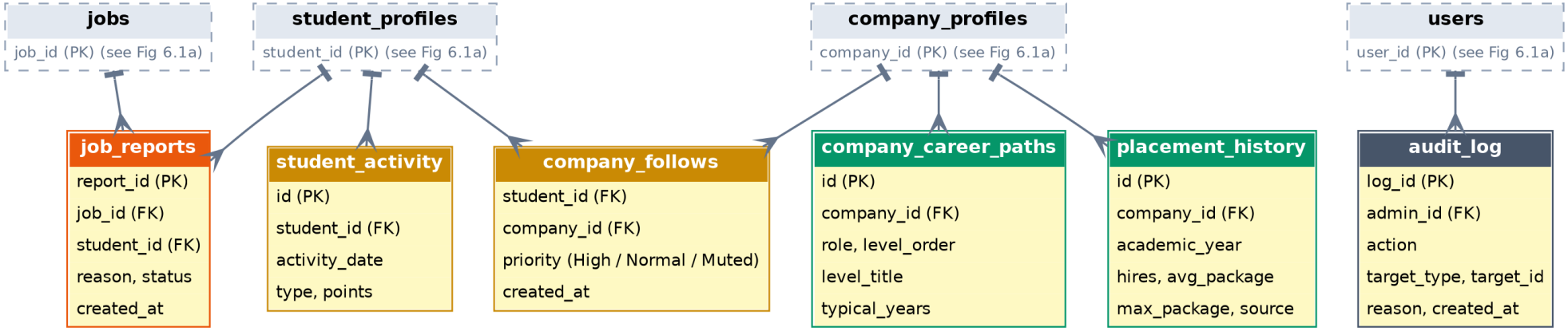


Figure 6.1c — Trust, hiring history, follow priority, engagement and audit log

Colour	Area
Grey	Accounts and roles, audit log
Blue	Student, CV and applications
Green	Company and jobs
Purple	Assessments and question bank
Orange	AI interview and job reports
Ochre	Notifications, consent, follow priority, engagement

6.2 Data Dictionary (Principal Entities)

Entity	Key attributes	Notes
users	user_id, email, password_hash, role_id, status, email_verified, created_at	Email unique; status is Active or Inactive only. A company's right to post comes from company_profiles.verification_status, not from users.status.
roles	role_id, name	STUDENT, COMPANY, ADMIN
student_profiles	student_id, user_id, full_name, college, degree, cgpa, graduation_year, preferred_roles	One per student user
company_profiles	company_id, user_id, company_name, industry, verification_status, contact_details, registration_no, website, official_email, trust_level, rejection_reason	verification_status: Pending / Verified / Rejected / Suspended. registration_no unique.
company_documents	doc_id, company_id, file_path, type, uploaded_at	Proof documents; file outside web root; visible to owner and administrators only
resumes	resume_id, student_id, file_path, extracted_text, uploaded_at	File on storage; metadata in MySQL
jobs	job_id, company_id, title, description, salary_min, salary_max, min_cgpa, deadline, work_mode, status	status: Draft / Pending_Review / Open / Closed / Hidden / Removed (Section 8.4)
job_skills	id, job_id, skill_name, is_required	Required or preferred skill
job_match_results	match_id, resume_id, job_id (nullable), source_type, jd_text, match_percent, ats_score, eligible	Indicative values. job_id is empty when a pasted job description is analysed; source_type: PASTED / VACANCY / PUBLISHED.
skill_gap_reports	report_id, match_id, missing_skills, suggestions	One per match result
applications	application_id, student_id, job_id, status, applied_at, offered_role, offered_package	Unique (student_id, job_id), so a withdrawn job cannot be re-applied. offered_* set only when Selected (LPA).
application_status_history	id, application_id, status, changed_by, changed_at	Append-only
question_bank	question_id, topic, company_tag, difficulty, question_text, options, answer	Managed by admin
tests	test_id, job_id, title, type, duration_minutes	Questions linked via test_questions
test_questions	test_id, question_id	Many-to-many link between tests and question_bank
test_attempts	attempt_id, test_id, student_id, score, submitted_at	Answers stored per attempt
interview_sessions	session_id, student_id, job_id (optional), mode, status	Mock or recruiter session
interview_feedback	feedback_id, session_id, question, answer, ai_feedback, score	AI output stored as recommendation
chat_history	chat_id, user_id, role, message, created_at	Per-user conversation
notifications	notification_id, user_id, message, type, priority, is_read, created_at	In-app notifications; priority NORMAL or HIGH
user_consents	consent_id, user_id, consent_type, granted_at	Stores AI-use consent (FR-CV-15)
company_follows	student_id, company_id, priority, created_at	Unique (student_id, company_id); priority High / Normal / Muted
placement_history	id, company_id, academic_year, hires, avg_package, max_package, source	source: PLATFORM / SELF_REPORTED; aggregated figures only
company_career_paths	id, company_id, role, level_order, level_title, typical_years	Shown after administrator approval

Entity	Key attributes	Notes
student_activity	id, student_id, activity_date, type, points	Unique (student_id, activity_date, type) for login; Later release
job_reports	report_id, job_id, student_id, reason, status, created_at	One report per student per job; status Open / Upheld / Dismissed
audit_log	log_id, admin_id, action, target_type, target_id, reason, created_at	Append-only (FR-ADM-07, FR-ADM-13)

6.3 Data Management Requirements

- **DR-01** Referential integrity shall be enforced with foreign keys between users, profiles, jobs, applications and assessment records.
- **DR-02** Uploaded CVs shall be kept until the student deletes them or the account is deleted; files shall be stored outside the public web root under non-guessable names.
- **DR-03** A user shall be able to request deletion of their account, CVs, chat history and interview data; the retention period for other records is to be decided (TBD-06).
- **DR-04** Application status history and audit logs shall be append-only.
- **DR-05** The database shall be backed up regularly in the deployment environment.
- **DR-06** Schema changes shall be applied through versioned migration scripts.
- **DR-07** (v1.1) Company verification documents shall be stored like CVs (outside the web root, non-guessable names) and be retained as long as the company account exists or the audit period (TBD-06) requires.
- **DR-08** (v1.1) Offered role and package shall be stored per application but exposed to students only as aggregates in placement_history (BR-14).
- **DR-09** (v1.1) Student activity and points data shall be readable only by the student and administrators and shall not be exposed to companies (BR-12).

7. Non-Functional Requirements

The numeric targets below are initial targets for the MVP. They shall be confirmed or adjusted during testing and are verified using the stated method.

7.1 Performance

ID	Requirement	Verification
NFR-PERF-01	Using a throttled Fast 4G profile (about 9 Mbps down, 1.6 Mbps up, 150 ms round-trip) with an empty cache, the Largest Contentful Paint (LCP) of the landing, login and dashboard pages shall be ≤ 3 seconds for 95% of runs. On an unthrottled connection of 10 Mbps or more it shall be ≤ 2 seconds.	Lighthouse CI or Chrome DevTools throttling, at least 20 runs per page
NFR-PERF-02	Non-AI API calls shall respond within 2 seconds for 95% of requests with up to 100 concurrent users.	Load test
NFR-PERF-03	Rule-based CV analysis (excluding the AI call) shall return its results within 5 seconds for a CV of up to 5 MB.	Timed test
NFR-PERF-04a	Any operation that can take longer than 5 seconds (AI suggestions, interview feedback, chatbot) shall run asynchronously. The API shall respond within 1 second, either with 202 Accepted and a task ID, or by opening a Server-Sent Events (SSE) stream.	Functional test
NFR-PERF-04b	Chatbot answers and interview feedback shall be streamed via SSE, with the first chunk arriving within 5 seconds for 95% of requests.	Functional test with throttled AI stub
NFR-PERF-04c	The hard limit for any AI operation shall be 30 seconds. After that the server shall send an error event, keep any partial output, and the UI shall offer a retry.	Fault-injection test (timeout)
NFR-PERF-04d	The UI shall never block. It shall show progress or a skeleton. For CV analysis the rule-based results (match %, skills, eligibility, ATS) shall appear first and AI suggestions load afterwards (consistent with FR-CV-14).	Functional test
NFR-PERF-04e	Task results shall be stored so that a page refresh or reconnect can retrieve them with GET /tasks/{id}.	Functional test
NFR-PERF-04f	Because the browser's native EventSource cannot send an Authorization header, the frontend shall use a fetch-based SSE client (for example @microsoft/fetch-event-source) or a short-lived one-time stream token.	Code review, security test
NFR-PERF-05	Timed-test submissions shall be evaluated and stored within 3 seconds.	Functional test

Figure D.3 in Appendix D shows the asynchronous request flow.

7.2 Security

ID	Requirement	Verification
NFR-SEC-01	Passwords shall be hashed with BCrypt; plain text passwords shall never be stored or logged.	Code review
NFR-SEC-02	JWTs shall expire (default 24 hours, configurable) and be validated on every protected request.	Security test
NFR-SEC-03	Role-based authorisation shall be enforced on the server for every endpoint; hidden UI controls shall not be the only protection.	Security test
NFR-SEC-04	Uploaded files shall be checked for type, size and content, and shall not be executable or served directly from the web root.	Security test
NFR-SEC-05	The system shall protect against SQL injection, cross-site scripting and cross-site request abuse through parameterised queries, input validation, output encoding and CORS restrictions.	Security test, code review
NFR-SEC-06	All production traffic shall use HTTPS.	Deployment check
NFR-SEC-07	CVs, assessment results and interview records shall be accessible only to their owner and to companies the student applied to, and to administrators where necessary.	Security test
NFR-SEC-08	Secrets (database password, JWT secret, AI API key, email credentials) shall be supplied through environment configuration, not stored in source code.	Code review

ID	Requirement	Verification
NFR-SEC-09	Repeated failed logins shall be rate-limited: after 5 failed attempts for the same email or IP address within 15 minutes, further attempts shall be blocked for 15 minutes (values configurable). Moved from Later to MVP.	Security test
NFR-SEC-10	Company verification documents shall be accessible only to the owning company and to administrators.	Security test
NFR-SEC-11	Task and SSE endpoints shall require authentication, verify that the task belongs to the requesting user and shall not place long-lived tokens in URLs.	Security test

7.3 Reliability and Availability

ID	Requirement	Verification
NFR-REL-01	A failure or timeout of the AI service shall not affect non-AI features.	Fault injection test
NFR-REL-02	Applications, test submissions and status changes shall be saved transactionally so partial data is not stored.	Integration test
NFR-REL-03	The system shall target 99% availability during a pilot period.	Monitoring
NFR-REL-04	Errors shall be logged with enough detail for diagnosis without exposing internals to users.	Inspection

7.4 Usability and Accessibility

ID	Requirement	Verification
NFR-USE-01	A new student shall be able to start a CV analysis from the dashboard within 3 clicks.	Usability test
NFR-USE-02	The UI shall be consistent across portals and responsive from 360 px to 1920 px.	Inspection, device test
NFR-USE-03	Form errors shall be shown in plain language next to the relevant field.	Inspection
NFR-USE-04	The UI shall aim to meet WCAG 2.1 AA basics: labelled inputs, sufficient contrast, keyboard navigation.	Accessibility check

7.5 Maintainability, Portability and Scalability

ID	Requirement	Verification
NFR-MNT-01	The backend shall be organised in feature modules with controller, service, repository, entity and DTO layers.	Code review
NFR-MNT-02	All REST endpoints shall be documented with OpenAPI (Swagger UI).	Inspection
NFR-MNT-03	Core services (matching, scoring, authentication) shall have automated tests, with a target of at least 60% line coverage.	Coverage report
NFR-MNT-04	Configuration shall be externalised so the same build runs in development and production.	Inspection
NFR-POR-01	The frontend shall work on the latest two versions of Chrome, Edge, Firefox and Safari.	Cross-browser test
NFR-SCL-01	The backend shall be stateless so it can be scaled horizontally; database queries shall use indexes on foreign keys and search columns. Async task state shall be kept in MySQL, not in server memory.	Design review

7.6 Privacy and Responsible AI

ID	Requirement	Verification
NFR-PRV-01	The system shall obtain the user's consent before sending CV or interview content to the external AI service (implemented by FR-CV-15).	Functional test
NFR-PRV-02	Only the data needed for the AI task shall be sent; passwords, tokens and unrelated personal data shall not be sent.	Code review
NFR-PRV-03	AI-generated content shall be labelled as a recommendation and shall not be presented as a guarantee of ATS result, eligibility or selection.	Inspection

ID	Requirement	Verification
NFR-PRV-04	Consequential recruitment decisions shall be made by humans; AI scores shall not trigger automatic rejection or selection.	Design review
NFR-PRV-05	Points, streaks and activity data shall not be visible to companies and shall not be used in matching, ranking or any hiring decision.	Design review, access-control test
NFR-PRV-06	Company hiring statistics shall be shown only as aggregates, never as individual offers, and hidden when fewer than 3 hires exist.	Functional test

8. Business Rules and Calculation Methods

8.1 Application Status Lifecycle

An application starts as Applied and moves through the statuses below. Selected, Rejected and Withdrawn are final. Every transition is stored in the status history. A withdrawn application cannot be resubmitted. Selecting a candidate requires the offered role and package.

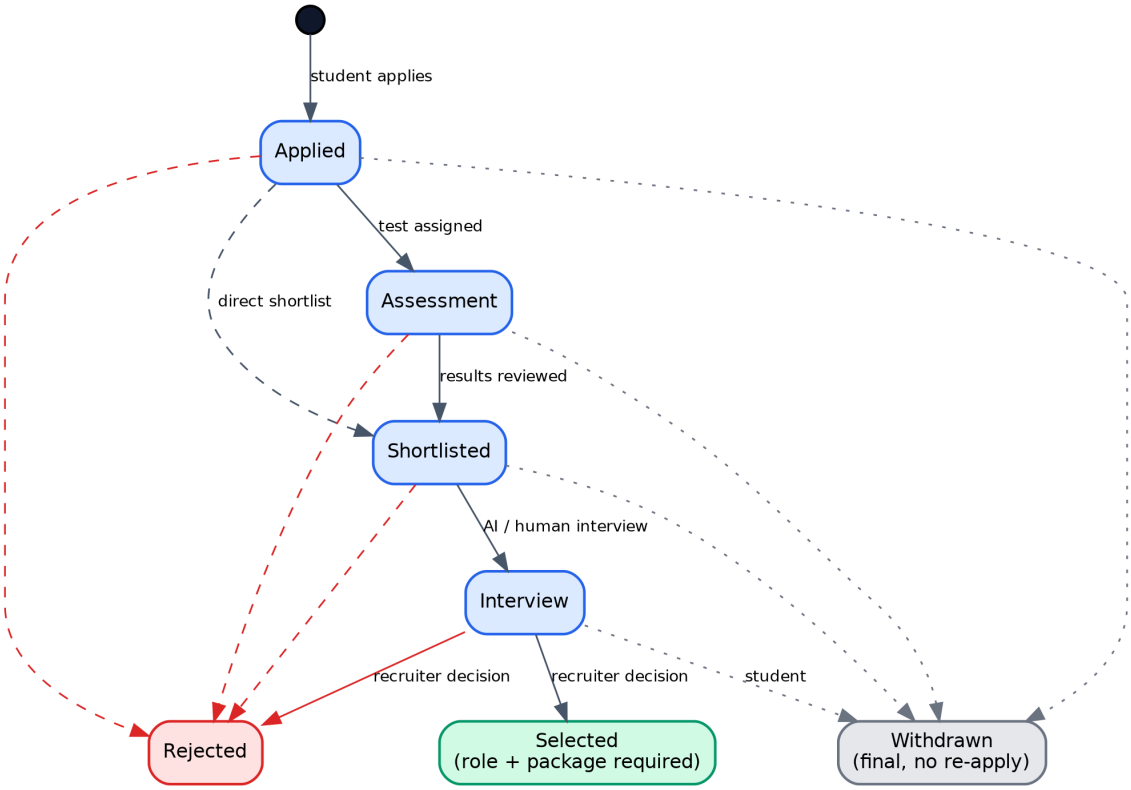


Figure 8.1 — Application status lifecycle

From	Allowed next status
Applied	Assessment, Shortlisted, Rejected, Withdrawn
Assessment	Shortlisted, Rejected, Withdrawn
Shortlisted	Interview, Rejected, Withdrawn
Interview	Selected, Rejected, Withdrawn
Selected / Rejected / Withdrawn	None (final)

8.2 Job-Match Calculation

The job-match percentage is rule-based and reproducible. The AI service explains gaps and suggests improvements but does not produce the number. The weights are proposed starting values and shall be configurable.

Component	Weight	Measurement
Required skills coverage	55%	Share of the job's required skills found in the CV (exact and synonym match). A job always has at least one required skill (FR-JOB-01).
Preferred skills coverage	20%	Share of the job's preferred skills found in the CV
Projects, certifications and experience relevance	25%	Keyword and role overlap between CV content and the job description

Match % = 0.55 × required coverage + 0.20 × preferred coverage + 0.25 × relevance score. Eligibility (CGPA, degree, graduation year) and the ATS readability result are reported separately and are not included in the percentage.

No preferred skills (v1.1): if a job lists no preferred skills, the 20% weight is redistributed so that Match % = 0.70 × required coverage + 0.30 × relevance score. This avoids a division by zero and keeps the maximum at 100%.

8.3 Business Rules

ID	Rule
BR-01	Only companies whose verification status is Verified may publish jobs. Pending, Rejected and Suspended companies cannot.
BR-02	A student may apply to a given job only once. A withdrawn application is final and cannot be resubmitted.
BR-03	Applications are not accepted after the deadline or after the job is closed.
BR-04	Only the company that owns a job may view its applicants and change application statuses.
BR-05	Status changes must follow the lifecycle in Section 8.1.
BR-06	ATS result, job-match percentage and eligibility are always shown as three separate results.
BR-07	CV uploads are limited to PDF (DOCX is Later) and a maximum of 5 MB.
BR-08	A timed test auto-submits when its time limit ends; answers given so far are scored.
BR-09	AI output is advisory; final hiring decisions are made by human recruiters.
BR-10	Closed jobs remain visible in the tracker of students who applied.
BR-11	Administrator accounts are created only by existing administrators or deployment seeding, never by public registration.
BR-12	Points and streaks are for student motivation only. They shall not be visible to companies and shall not affect match %, ranking or any hiring decision.
BR-13	Jobs that ask applicants for payment are prohibited and shall be removed.
BR-14	Company hiring statistics are shown only as aggregates and only when at least 3 hires exist for the period; individual offers are never shown to students.

8.4 Job Status Rules

From	Allowed next status	Trigger
Draft	Open, Pending_Review	Verified company publishes. Pending_Review if it is one of the first 3 jobs or is flagged by the scam screening.
Pending_Review	Open, Removed	Administrator approves or rejects (reason logged).
Open	Closed, Hidden, Removed	Company closes or deadline passes (Closed); administrator or report threshold hides (Hidden); administrator removes (Removed).
Hidden	Open, Removed	Administrator restores or removes after review.
Closed / Removed	None (final)	A new job must be created. When a company is suspended, all its Open jobs become Hidden.

8.5 Engagement Points (Later)

Points reward meaningful activity, not logins alone, so that students cannot gain many points just by logging in. A daily cap applies (default 50, configurable). A streak may skip one day per week without breaking. Final values are open (TBD-09).

Activity	Points
Daily login (once per day)	5
CV analysis completed	10
Practice test completed	10, plus up to 5 bonus for score
Mock interview completed	15
Job application submitted	5
Profile 100% complete (one time)	20

Activity	Points
7-day streak bonus	25

8.6 Initial Scam Indicators (FR-JOB-10)

- Payment requests: "registration fee", "security deposit", "pay to apply", "training fee", "refundable deposit".
- Unrealistic promises: "guaranteed job", "100% placement", very high salary for no skills or experience.
- Contact only through WhatsApp or Telegram, or a personal email address (for example Gmail) instead of a company domain.
- The list is configurable and is tuned with sample postings (TBD-10). A flag never removes a job by itself; an administrator decides.

Appendix A. Requirements Traceability Matrix

The matrix links each feature group to its requirements, use cases, backend module, MVP development phase and verification approach. All IDs use the standard hyphenated form (the broken IDs of v1.0 are corrected).

Feature	Requirements	Use cases	Module	Phase	Verification
Authentication and users	FR-AUTH-01–09	UC-01, UC-02	auth, user	1	API and security tests
Student profile	FR-STU-01–06	UC-01	student	2	Functional test
Jobs and browsing	FR-JOB-01–08	UC-03	job, company	2	Functional test
Job protection (fake jobs)	FR-JOB-09–13	UC-03, UC-12, UC-13	job, trust-safety	2	Functional and security test
Applications	FR-APP-01–09	UC-06	application	2	Integration test
Applicant management	FR-CAM-01–06	UC-07	application, company	2	Access-control test
Company profile and hiring history	FR-COM-01–08	UC-07	company, analytics	2 / 5	Functional and privacy test
CV Analyzer	FR-CV-01–15	UC-04, UC-05	cv	3	Unit and functional test
Assessments and practice	FR-TST-01–10	UC-08	assessment	4	Functional test
AI interview	FR-INT-01–09	UC-09	ai	4	Functional, fault test
AI chatbot	FR-BOT-01–05	UC-10	ai	4	Functional, fault test
Notifications, alerts and follow priority	FR-NOT-01–10, FR-REC-01–03	UC-03, UC-06, UC-11	notification	2	Functional test
Dashboards and engagement	FR-DSH-01–09	—	analytics, engagement	5	Functional test
Administration and trust	FR-ADM-01–15	UC-02, UC-13	admin, trust-safety	1 / 2 / 5	Functional and access-control test
User interface	FR-UI-01–05	All	frontend	1–5	Inspection, device test
Async AI and streaming	NFR-PERF-04a–f, FR-UI-04	UC-04, UC-09, UC-10	async-tasks, ai	3 / 4	Functional and fault-injection test

Appendix B. MVP Acceptance Criteria

The MVP is accepted when all criteria below pass.

ID	Acceptance test	Requirements
AC-01	Given valid details, a student and a company can register and log in and each sees only their own portal.	FR-AUTH-01 to 05
AC-02	A verified company can publish a job with a complete description, skills, eligibility, salary and deadline; a Pending, Rejected or Suspended company cannot.	FR-JOB-01, 03, 05
AC-03	A student can upload a PDF CV and receive a CV-to-job analysis report.	FR-CV-01 to 03, 05 to 09, 15
AC-04	A student can compare one CV with multiple published jobs and see a separate result for each.	FR-CV-04, 10
AC-05	Reports show matching skills, missing skills and recommendations.	FR-CV-06, 09
AC-06	A student can apply for a job and track its status through changes; a withdrawn application cannot be resubmitted.	FR-APP-01 to 07, BR-02
AC-07	A company can review applicants, view test results and shortlist candidates.	FR-CAM-01 to 04, FR-TST-06

ID	Acceptance test	Requirements
AC-08	The chatbot and mock interview work when the AI service is available and show a clear error when it is not.	FR-BOT-01 to 04, FR-INT-01 to 06
AC-09	Application, job, profile and assessment data are stored and retrieved correctly from MySQL.	DR-01, FR-TST-04
AC-10	Unauthorised users cannot access restricted student or company data or company verification documents.	FR-AUTH-05, FR-CAM-05, NFR-SEC-03, 07, 10
AC-11	Fake-company and fake-job protection works: registration without registration number or proof is rejected; an administrator cannot approve before completing the checklist; a duplicate registration number raises a warning; the first 3 jobs of a new company are held as Pending_Review; a job containing "registration fee" goes to the review queue; 3 student reports hide a job; suspending a company hides all its jobs and notifies applicants.	FR-ADM-08 to 13, FR-ADM-15, FR-JOB-09 to 11, FR-JOB-13
AC-12	A student can follow a company and set High, Normal or Muted. A new job from a High-priority company gives a Priority notification listed first; Muted gives none; an 11th High-priority company is refused when the limit is 10; no duplicate notification is created for one job.	FR-NOT-04 to 06, 08, 10
AC-13	Setting Selected without role and package is refused. After Selected, the company profile shows per-year aggregates labelled Platform-verified; with fewer than 3 hires averages are hidden; no individual offer is visible to students.	FR-COM-04, 05, 07, FR-CAM-03
AC-14	A CV analysis shows rule-based results within 5 seconds while AI suggestions load afterwards; chatbot streaming starts within 5 seconds; at the 30-second limit an error with retry appears and partial output is kept; after a page refresh the result is retrieved by task ID; AI is blocked until consent is given.	NFR-PERF-03, 04a–f, FR-CV-09, 14, 15, FR-BOT-04, FR-INT-06
AC-15	The 6th failed login within the rate-limit window is blocked, and every administrator verification, rejection, suspension and report decision appears in the audit log.	NFR-SEC-09, FR-ADM-07, FR-ADM-13

Appendix C. Open Issues (To Be Decided)

ID	Issue	Decision needed	Status
TBD-01	AI provider and model	Choose provider, cost limit and rate limits	Open
TBD-02	Hosting and deployment environment	Choose host, domain and backup method	Open
TBD-03	Email provider (SMTP)	Needed in the MVP for company email verification (FR-ADM-09); password reset and other emails stay Later. Choose service.	Open (changed)
TBD-04	Final matching weights and skill synonym list	Tune using sample CVs and jobs	Open
TBD-05	DOCX CV support in the first release	Decided: PDF only in the MVP; DOCX is Later.	Resolved in v1.1
TBD-06	Data retention period for applications, assessments, documents and logs	Agree with institution or policy	Open
TBD-07	Standard list of degrees and eligibility fields	Define dropdown values for job and profile forms	Open
TBD-08	Registration-number types to accept and validate	Decide which of CIN, GSTIN or others are accepted and what format checks to apply	Open (new)
TBD-09	Final point values and streak grace rule	Tune with student feedback (Later release)	Open (new)
TBD-10	Scam indicator list and thresholds	Confirm keyword list, first-3-jobs rule, report threshold of 3 and daily posting limit	Open (new)

Appendix D. Supporting Diagrams

D.1 End-to-End Placement Workflow

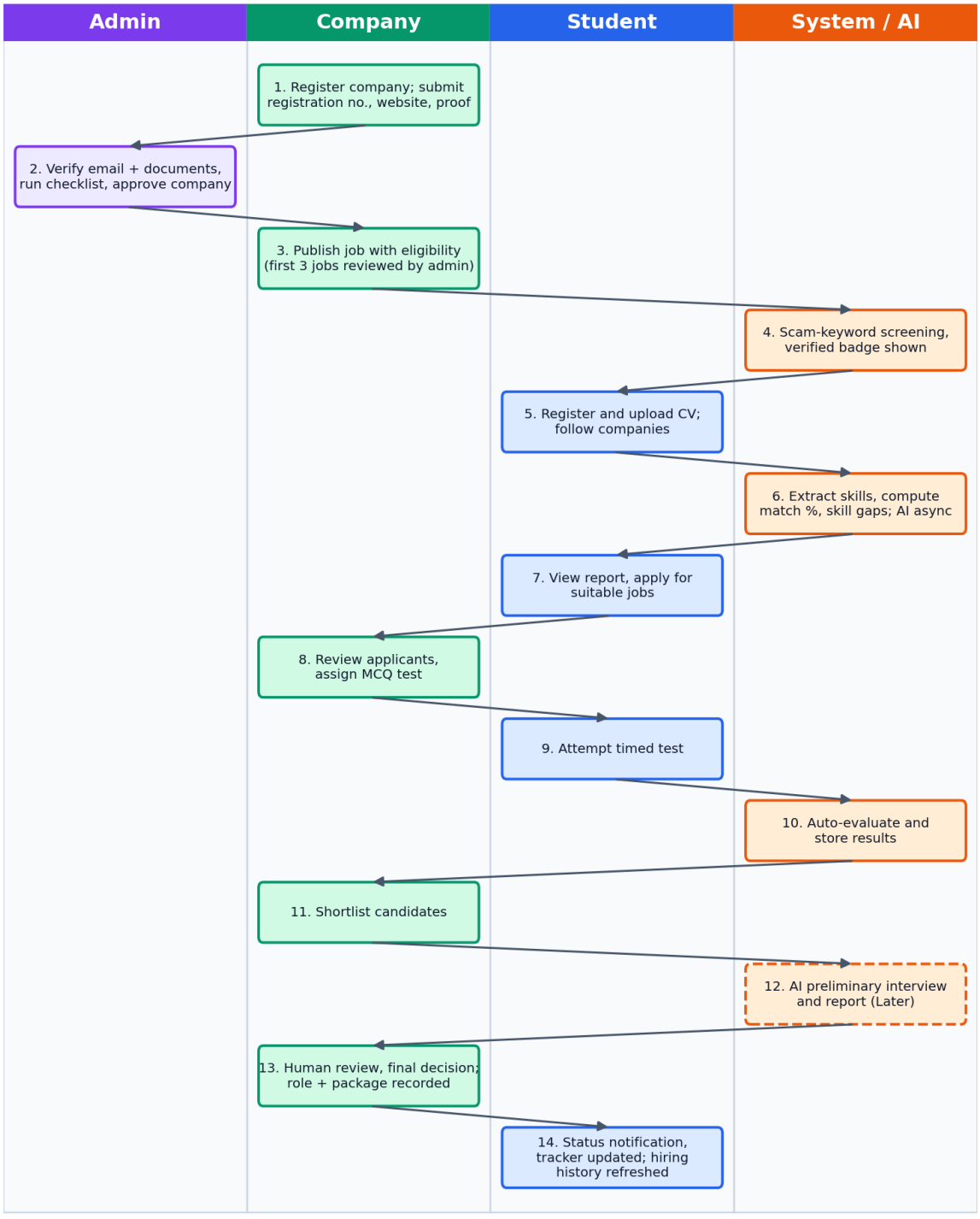


Figure D.1 — End-to-end workflow (swimlane)

D.2 Smart CV Analyzer Processing Flow

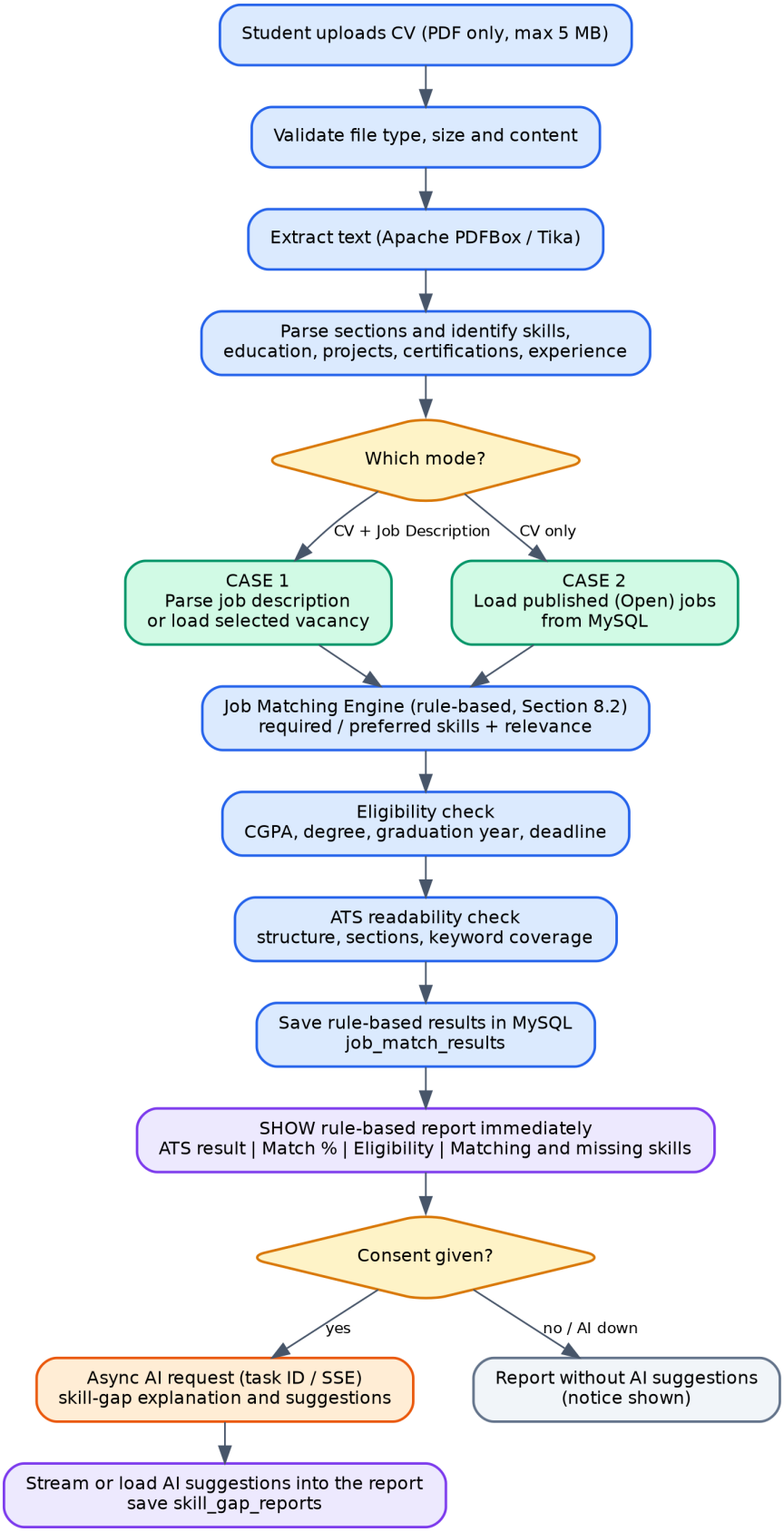


Figure D.2 — CV analysis and job-matching pipeline (PDF only; rule-based results first, AI suggestions asynchronous)

D.3 Asynchronous AI Request Flow

Any operation expected to take longer than 5 seconds returns immediately and delivers its result by SSE or by polling the task (NFR-PERF-04a to 04f).

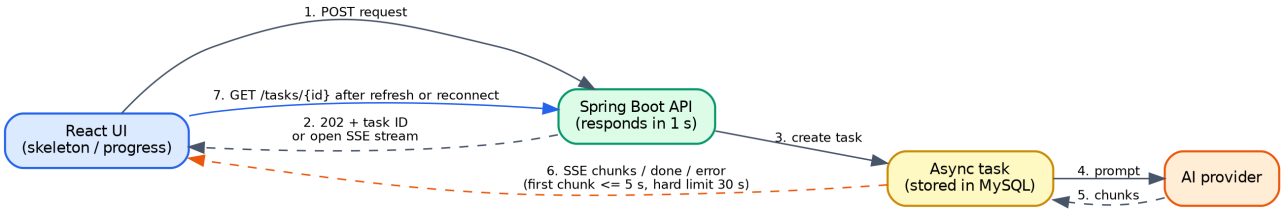


Figure D.3 — Asynchronous AI request with Server-Sent Events

D.4 Company and Job Verification Flow

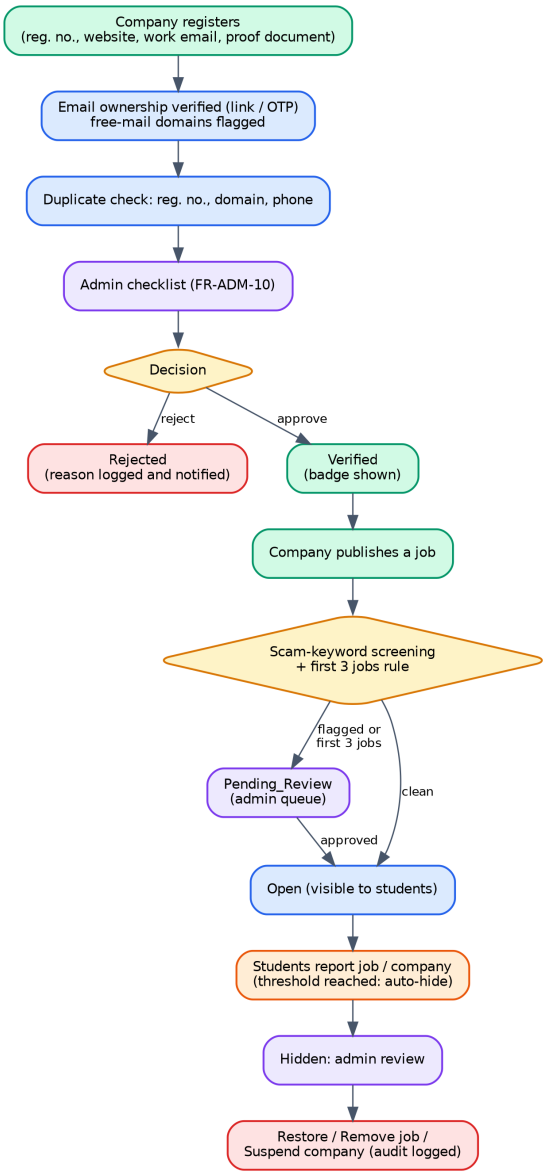


Figure D.4 — Layered protection against fake companies and fake jobs

Appendix E. Change Log from v1.0 to v1.1

E.1 Consistency fixes

Ref	Change	Where
C-01	Contradiction resolved: BR-02 allowed resubmission after withdrawal, but FR-APP-01, the unique (student_id, job_id) constraint and Withdrawn being final said otherwise. Withdrawal is now final.	BR-02, FR-APP-01, FR-APP-06, 8.1, UC-06
C-02	Broken IDs corrected (UC02, UC05, FRTST-06, FRINT-01, FRCAM-05 and similar).	Appendix A, Appendix B
C-03	Match % formula had no rule for jobs without preferred skills; weights now 70% required and 30% relevance in that case.	8.2
C-04	job_match_results.job_id made nullable; jd_text and source_type added for pasted job descriptions (Case 1).	6.2, Fig 6.1a
C-05	Job statuses extended with Pending_Review, Hidden and Removed so that FR-JOB-08 can be implemented; status rules added.	6.2, 8.4, FR-JOB-08
C-06	test_questions table added to the ER diagram and data dictionary.	Fig 6.1b, 6.2
C-07	Overlap of users.status and company verification_status removed: users.status is Active / Inactive; company states are Pending / Verified / Rejected / Suspended.	6.2, UC-01
C-08	PDF-only CV decision (TBD-05 resolved); Figure D.2 and BR-07 updated.	FR-STU-04, FR-CV-01, BR-07, 2.5, Fig D.2
C-09	NFR-SEC-09 (login rate limiting) and FR-ADM-07 (audit log) moved from Later to MVP.	7.2, 4.14
C-10	Duplicate dark-mode requirements merged into FR-UI-05; FR-STU-06 withdrawn.	3.1, 4.2
C-11	FR-CV-15 added so that AI consent (NFR-PRV-01) has a functional requirement and a data table.	4.4, 6.2
C-12	UC-01 alternate flows restated as a Step / Condition / System response table.	UC-01
C-13	Email service is needed in the MVP for company email verification, although TBD-03 said email was Later. Interface, architecture and TBD-03 updated.	3.3, Fig 2.1, TBD-03
C-14	FR-NOT-10 added to avoid duplicate notifications between matching-job alerts and followed-company alerts.	4.13
C-15	All figures redrawn as high-resolution diagrams; use case figures sized for reading; ER diagram on a landscape page.	All figures

E.2 New and changed features

Ref	Change	Where
N-01	Company hiring history per academic year, platform-verified and aggregated (minimum 3 hires), plus career growth paths (Later).	FR-COM-04 to 08, BR-14, DR-08, NFR-PRV-06
N-02	Student activity days, streaks and points with a daily cap (Later), kept private from companies.	FR-DSH-06 to 09, 8.5, BR-12, NFR-PRV-05
N-03	Follow companies with High / Normal / Muted priority and priority notifications, deadline reminders and limits.	FR-NOT-04 to 10, UC-11
N-04	Fake company protection: registration number, work email, proof document, email verification, duplicate detection, admin checklist, suspension, audit log.	FR-ADM-08 to 15, UC-02, UC-13, BR-01
N-05	Fake job protection: first-3-jobs review, scam screening, student reports with auto-hide, verified badge, payment warning, posting limit (Later).	FR-JOB-09 to 13, BR-13, UC-03, UC-12, 8.4, 8.6
N-06	NFR-PERF-01 rewritten with a measurable network profile and LCP metric; NFR-PERF-03 target set to 5 s; NFR-PERF-04 replaced by the asynchronous / SSE set 04a–f.	7.1, FR-UI-04, FR-INT-04, FR-INT-06, FR-BOT-01, FR-BOT-04, UC-04, Fig D.2, Fig D.3

Ref	Change	Where
N-07	New acceptance criteria AC-11 to AC-15, changed AC-02, AC-03, AC-06, AC-10; open issues TBD-08 to TBD-10 added; TBD-03 and TBD-05 updated; traceability matrix extended.	Appendices A, B, C